

V3 empirical study · Instruction

Six models · three underlyings · four volatility regimes · 1.5-year monthly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of six models tracks realized stock price S_t most closely, and does the ranking change across companies and the four volatility regimes?
- This report does not price options. Option RMSE lives in the companion decision PDF.
- In every table the BEST row is the lowest percentage RMSE of the p50 path versus realized S_t . That row is bold and shaded green; Mark ranks 1-6.

Experimental design (held fixed for every model)

- Models: GBM, GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH-Merton, Heston-Merton (Bates).
- Companies: SPY, AAPL, MSFT. Each name is calibrated and simulated separately.
- Regimes: Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly. No look-ahead.
- P-measure: keep the lookback drift μ . Do not replace $\mu \rightarrow r$ (that replacement is only for LSM option pricing).

V3 empirical study · Instruction (continued)

Six models · three underlyings · four volatility regimes · 1.5-year monthly calibration

Path construction

- One rolling path cloud per ticker $\times$ regime $\times$ model, $n_paths = 10000$, seed 42.
- Clock: daily trading days from the first bar of the regime through the regime end. $\Delta t = 1/252$.
- Start at the observed open S_0 . Refresh parameters at month-ends with data $\leq$ that date.
- At each day, summarize the cloud by p10, p25, p50, p75, p90. The reported path is p50, not the Monte Carlo mean.

Metrics

- Primary: $RMSE\% = 100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. Ranking uses this only.
- $MAE = \text{mean } |p50 - S_t|$. Bias \$ = $\text{mean}(p50 - S_t)$. Bias% = $100 \times \text{mean}((p50 - S_t)/S_t)$.
- ICP 25–75 = share of S_t inside $[p25, p75]$. ABW = $\text{mean}(p75 - p25)$. ICP 10–90 is recorded but not a table column.
- ICP is higher-better and is not used for the green row.

Page 3 · Table block · SPY · six models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	25.80	17.4631	+16.2305	+19.23	39.3%	26.0805	1
GARCH	34.61	25.7252	+25.2517	+29.51	19.8%	25.4872	6
Heston	29.06	20.1638	+19.5533	+22.97	41.3%	30.4680	4
Merton	27.61	19.2509	+18.4843	+21.84	36.5%	26.2971	2
GARCH-Merton	34.56	25.6850	+25.2189	+29.47	23.0%	26.6347	5
Heston-Merton	28.59	19.7154	+19.0528	+22.41	40.9%	29.0026	3

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	4.49	7.2337	+7.2240	+3.72	85.0%	20.8529	1
GARCH	6.45	10.8508	+10.8471	+5.55	54.5%	21.1444	5
Heston	6.37	11.0902	+11.0902	+5.70	84.6%	25.6579	3
Merton	7.02	11.6343	+11.6296	+5.94	47.4%	21.1555	6
GARCH-Merton	6.38	10.7511	+10.7486	+5.51	63.6%	22.2041	4
Heston-Merton	5.93	10.3272	+10.3272	+5.31	88.1%	26.6193	2

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	8.90	21.0394	+21.0394	+7.69	47.8%	36.9345	1
GARCH	19.54	51.9927	+51.9927	+18.42	1.2%	39.5574	5
Heston	10.70	26.7910	+26.7910	+9.72	55.0%	47.9049	3
Merton	13.33	35.1454	+35.1454	+12.63	1.2%	38.2732	4
GARCH-Merton	19.59	52.1721	+52.1721	+18.49	1.2%	43.7182	6
Heston-Merton	10.61	26.6338	+26.6338	+9.66	58.2%	50.1318	2

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	8.57	20.0495	-7.7461	-1.91	55.6%	46.6240	2
GARCH	15.40	33.0268	+27.2883	+9.48	46.0%	48.5569	6
Heston	8.53	16.7104	+0.6396	+0.82	70.2%	56.8537	1
Merton	9.02	18.3611	-0.1484	+0.60	60.3%	46.3915	4
GARCH-Merton	15.26	32.6217	+27.0683	+9.40	46.8%	52.5757	5
Heston-Merton	9.01	16.7711	+2.5418	+1.47	80.6%	62.5078	3

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 = share of S_t inside [p25, p75]. ABW = mean(p75-p25). Ranking uses RMSE% only.

Page 4 · Table block · AAPL · six models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	51.72	1.2223	+1.0434	+37.11	38.1%	1.9491	1
GARCH	92.20	2.6519	+2.5997	+81.91	5.2%	2.7575	6
Heston	54.05	1.2738	+1.1028	+39.04	40.9%	2.4853	3
Merton	56.69	1.3367	+1.2170	+42.41	36.5%	2.0099	4
GARCH-Merton	91.34	2.6155	+2.5641	+80.98	6.0%	2.9156	5
Heston-Merton	53.22	1.2553	+1.0730	+38.13	42.1%	2.4248	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	18.03	3.3957	-3.2014	-14.06	26.9%	4.4230	6
GARCH	13.35	2.5902	-2.1492	-9.14	43.9%	4.5854	2
Heston	17.49	3.2991	-3.0701	-13.42	26.1%	4.2802	4
Merton	11.21	2.1896	-1.6421	-6.75	74.7%	4.8326	1
GARCH-Merton	13.95	2.7057	-2.2555	-9.60	59.3%	5.3681	3
Heston-Merton	17.65	3.3285	-3.1026	-13.57	25.7%	3.8796	5

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	30.71	11.8415	+11.8300	+27.12	13.5%	13.1409	2
GARCH	44.15	18.6766	+18.6654	+41.28	1.6%	15.2479	6
Heston	32.95	12.8948	+12.8840	+29.44	18.3%	15.6787	4
Merton	29.96	11.4880	+11.4750	+26.34	12.7%	12.8160	1
GARCH-Merton	43.97	18.5901	+18.5790	+41.11	1.6%	16.9782	5
Heston-Merton	32.23	12.4873	+12.4757	+28.57	16.7%	14.7949	3

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	26.10	19.2567	-19.2474	-23.41	8.7%	15.5628	6
GARCH	11.19	6.9756	-2.9108	-3.35	56.7%	22.2427	2
Heston	24.17	17.6986	-17.6654	-21.31	10.7%	16.2181	3
Merton	24.55	18.0488	-18.0194	-21.87	11.5%	15.8998	5
GARCH-Merton	10.94	6.7900	-3.0443	-3.52	62.7%	24.5315	1
Heston-Merton	24.44	17.8804	-17.8476	-21.51	10.3%	15.6759	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 = share of S_t inside [p25, p75]. ABW = mean(p75-p25). Ranking uses RMSE% only.

Page 5 · Table block · MSFT · six models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	18.26	2.3103	+0.7057	+6.51	52.0%	5.1013	1
GARCH	21.87	2.5521	+1.8021	+13.72	48.0%	6.2418	6
Heston	19.59	2.4457	+1.0186	+8.59	54.0%	5.7817	4
Merton	18.66	2.4237	+0.3798	+4.33	40.5%	4.8143	2
GARCH-Merton	20.94	2.4827	+1.6051	+12.42	52.0%	6.8600	5
Heston-Merton	19.24	2.3987	+0.9634	+8.21	53.6%	5.5905	3

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	4.49	1.4435	-1.2021	-3.08	95.7%	7.6215	4
GARCH	9.17	2.6689	+2.5857	+6.72	87.7%	9.5330	6
Heston	3.82	1.1944	-0.9156	-2.33	95.3%	6.5944	3
Merton	2.73	0.7663	-0.0480	-0.04	94.1%	7.6709	1
GARCH-Merton	8.47	2.4706	+2.3935	+6.24	94.5%	10.7365	5
Heston-Merton	3.67	1.1428	-0.8290	-2.09	94.1%	6.1190	2

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	12.44	11.7781	+11.7781	+10.84	54.2%	27.6745	2
GARCH	22.51	24.9345	+24.9345	+21.58	1.2%	30.4957	6
Heston	13.84	13.6959	+13.6959	+12.47	53.8%	29.2705	4
Merton	12.29	10.8532	+10.8532	+10.16	55.0%	26.9703	1
GARCH-Merton	22.00	24.3351	+24.3351	+21.09	1.2%	33.5039	5
Heston-Merton	13.56	13.1550	+13.1550	+12.04	53.0%	27.7019	3

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	10.72	15.8101	-14.2517	-7.67	79.0%	36.7213	5
GARCH	5.92	6.6186	-2.1890	-0.90	90.1%	37.3418	2
Heston	9.24	13.2810	-10.7368	-5.56	89.7%	40.0924	4
Merton	10.88	16.2008	-15.0037	-8.18	77.8%	36.7996	6
GARCH-Merton	5.85	6.5269	-2.2403	-0.93	90.5%	39.6506	1
Heston-Merton	9.15	12.9707	-10.1693	-5.19	86.5%	39.8079	3

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 = share of S_t inside [p25, p75]. ABW = mean(p75-p25). Ranking uses RMSE% only.

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 12 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 12 cells best
1	GBM	18.35	15.24	5
2	Merton	18.66	12.81	4
3	Heston–Merton	18.94	15.60	0
4	Heston	19.15	15.66	1
5	GARCH–Merton	24.44	17.43	2
6	GARCH	24.70	17.47	0

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	Total
GBM	3	1	1	5
GARCH	0	0	0	0
Heston	1	0	0	1
Merton	0	2	2	4
GARCH–Merton	0	1	1	2
Heston–Merton	0	0	0	0

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	31.92	9.01	17.35	15.13	18.35
GARCH	49.56	9.66	28.73	10.84	24.70
Heston	34.24	9.23	19.16	13.98	19.15
Merton	34.32	6.99	18.53	14.82	18.66
GARCH–Merton	48.95	9.60	28.52	10.68	24.44
Heston–Merton	33.68	9.08	18.80	14.20	18.94

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes bes
GBM	11.94	8.73	3
GARCH	19.00	17.47	0
Heston	13.67	9.61	1
Merton	14.24	11.17	0
GARCH-Merton	18.95	17.43	0
Heston-Merton	13.54	9.81	0

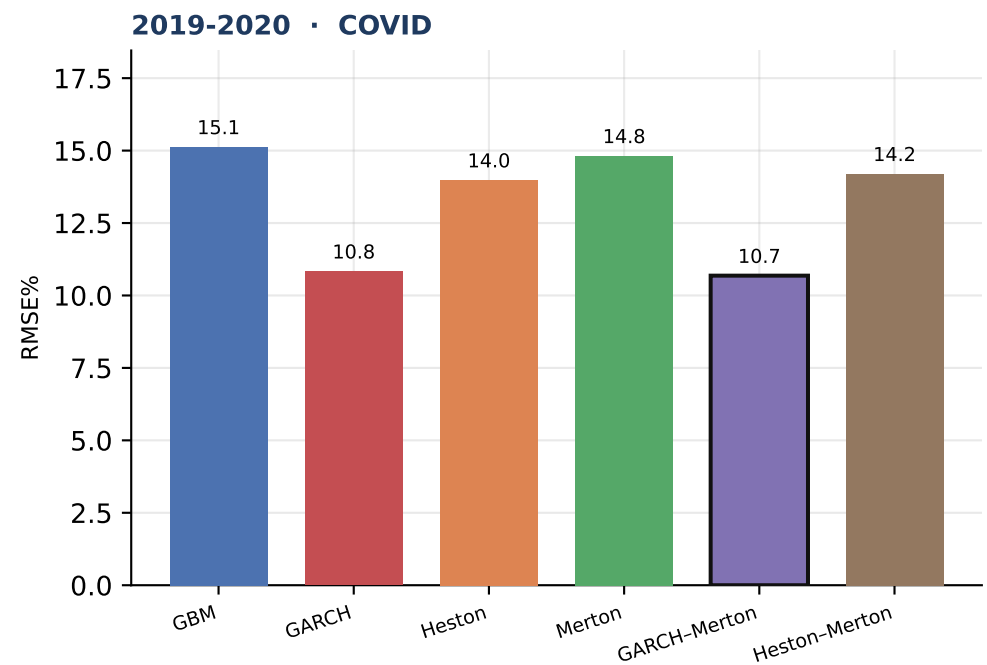
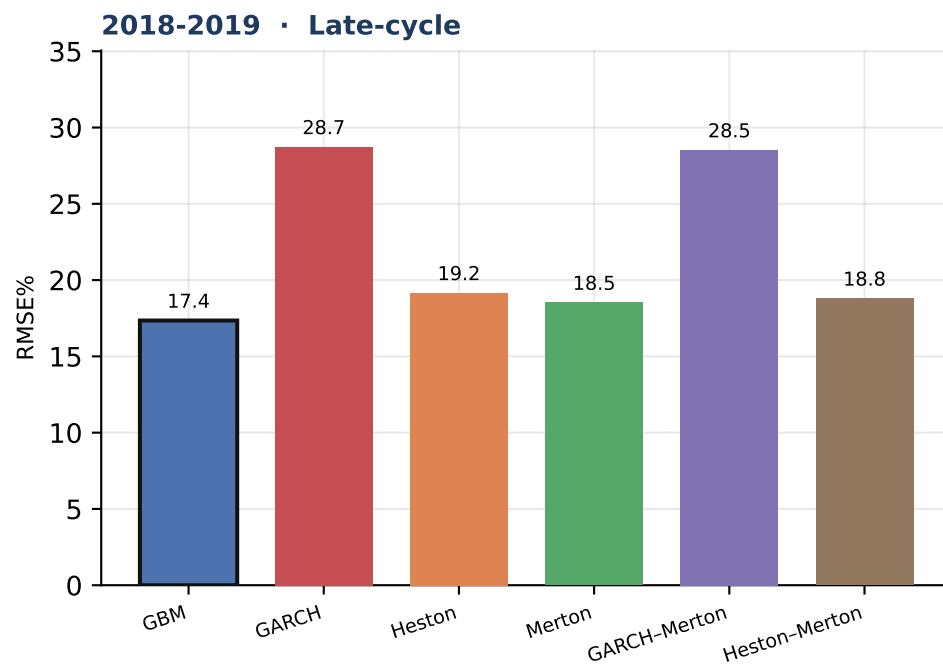
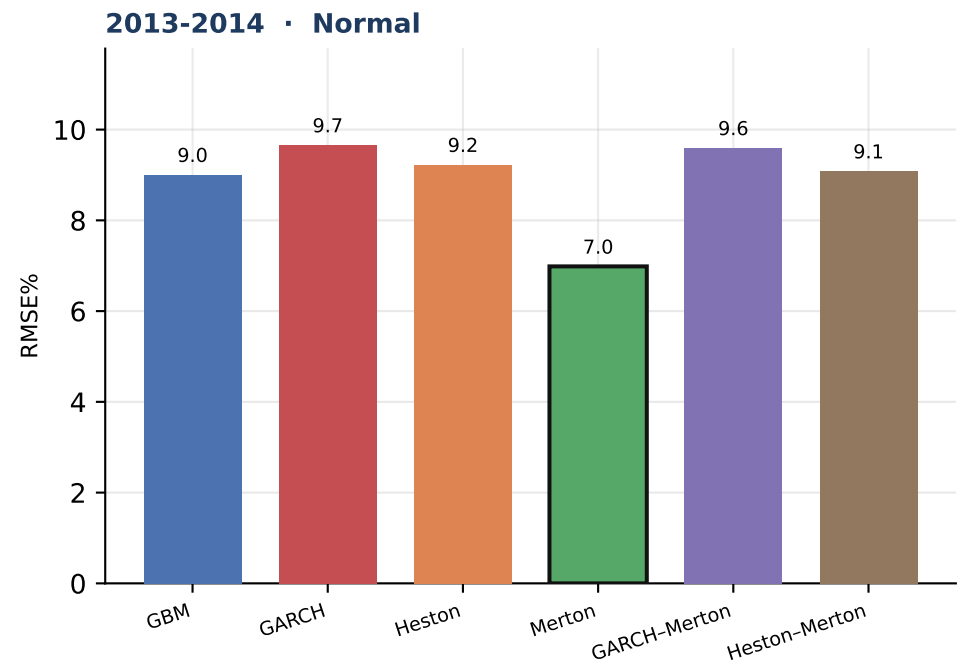
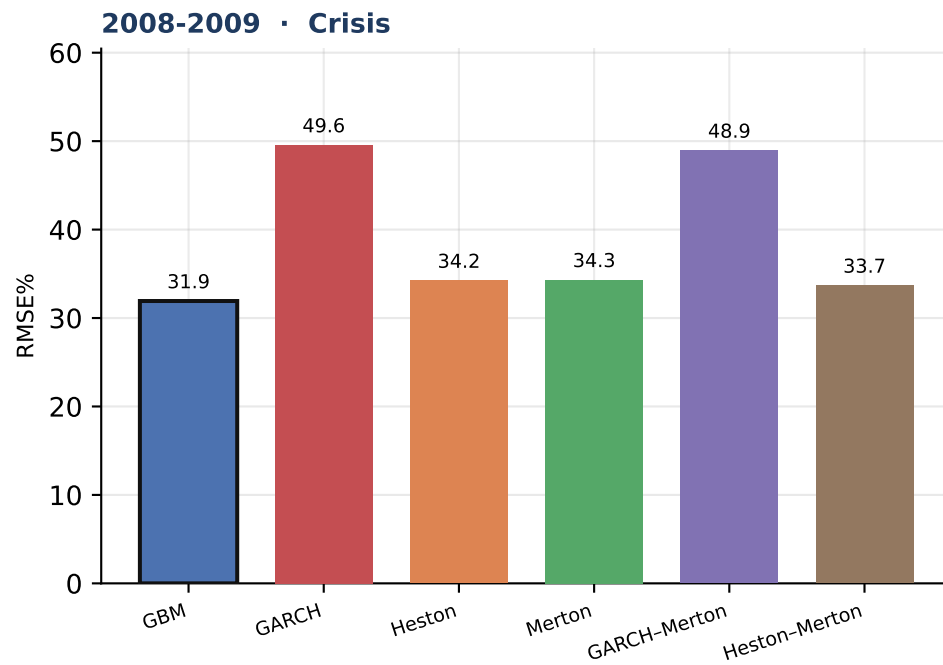
Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes bes
GBM	31.64	28.41	1
GARCH	40.22	28.75	0
Heston	32.17	28.56	0
Merton	30.60	27.26	2
GARCH-Merton	40.05	28.96	1
Heston-Merton	31.88	28.33	0

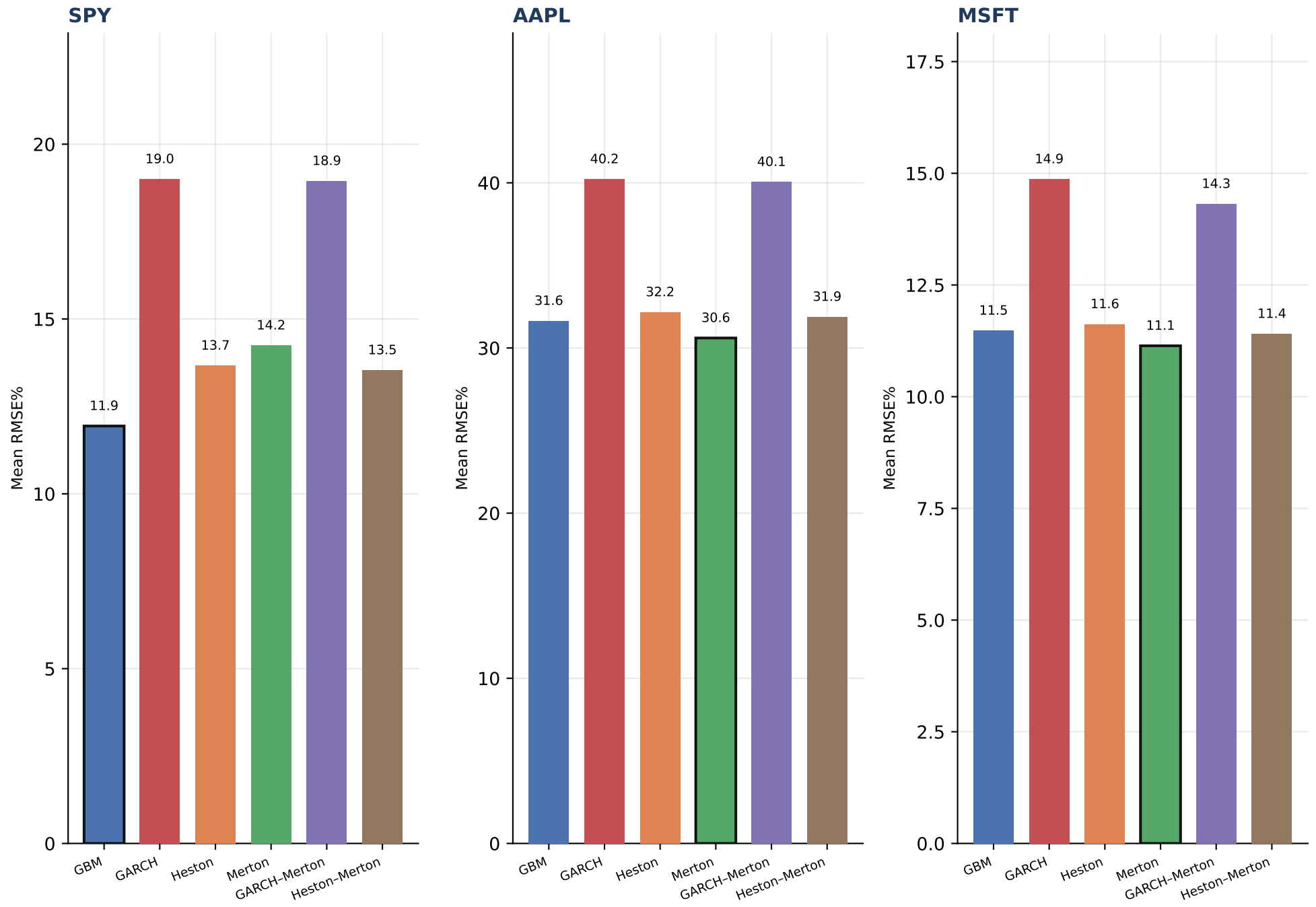
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes bes
GBM	11.48	11.58	1
GARCH	14.87	15.52	0
Heston	11.62	11.54	0
Merton	11.14	11.58	2
GARCH-Merton	14.31	14.70	1
Heston-Merton	11.41	11.35	0

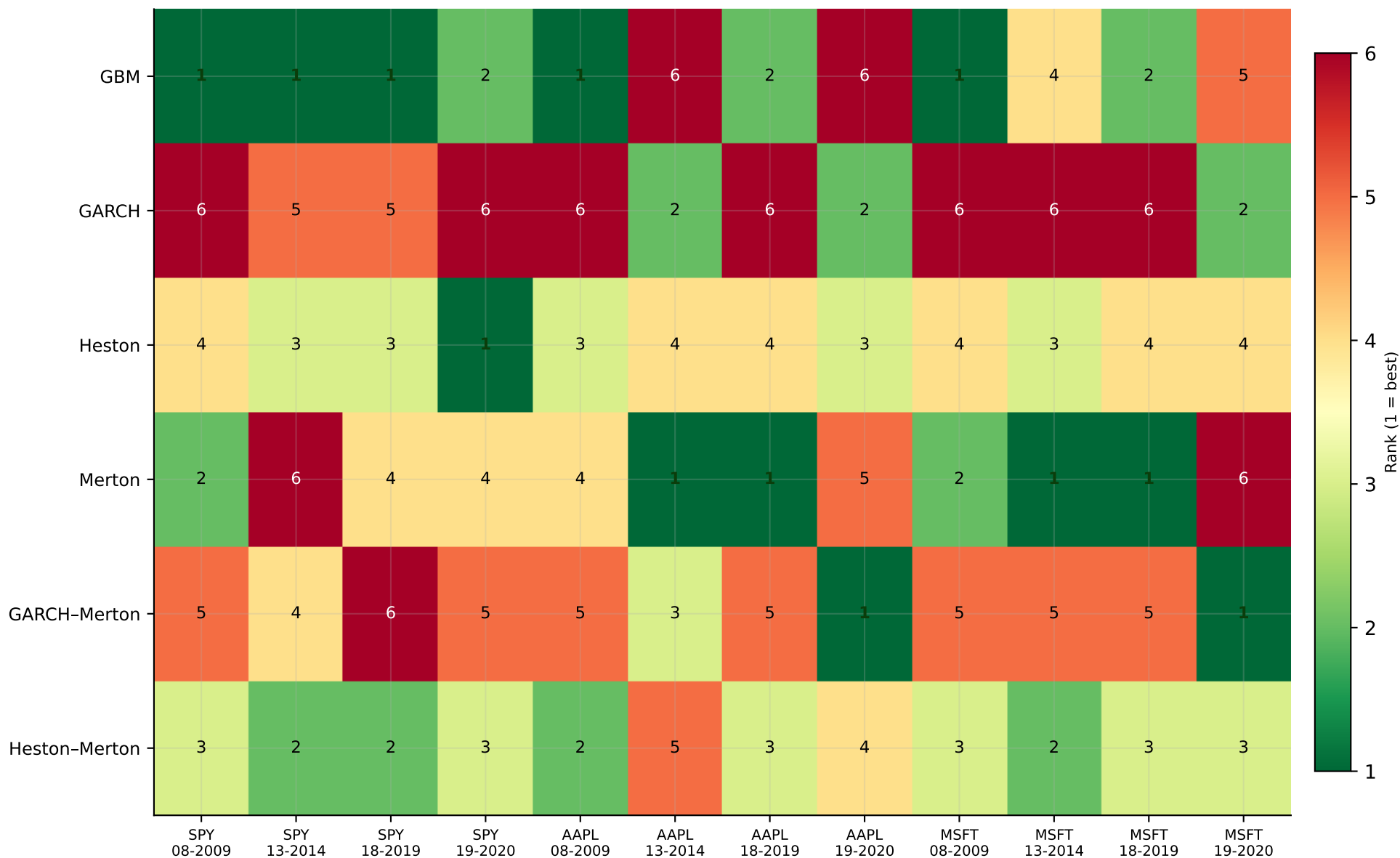
Each column is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



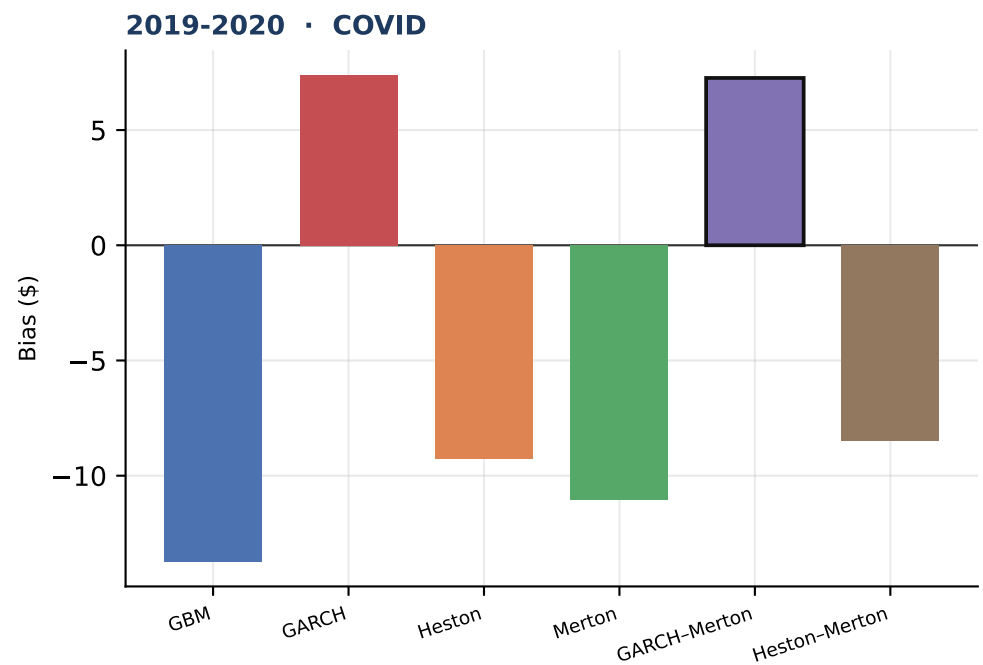
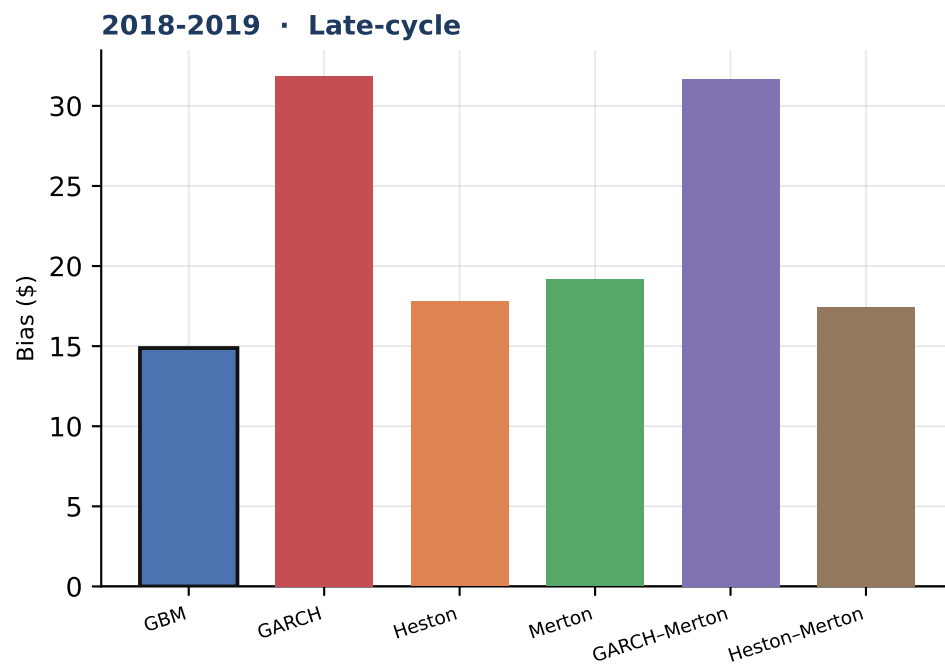
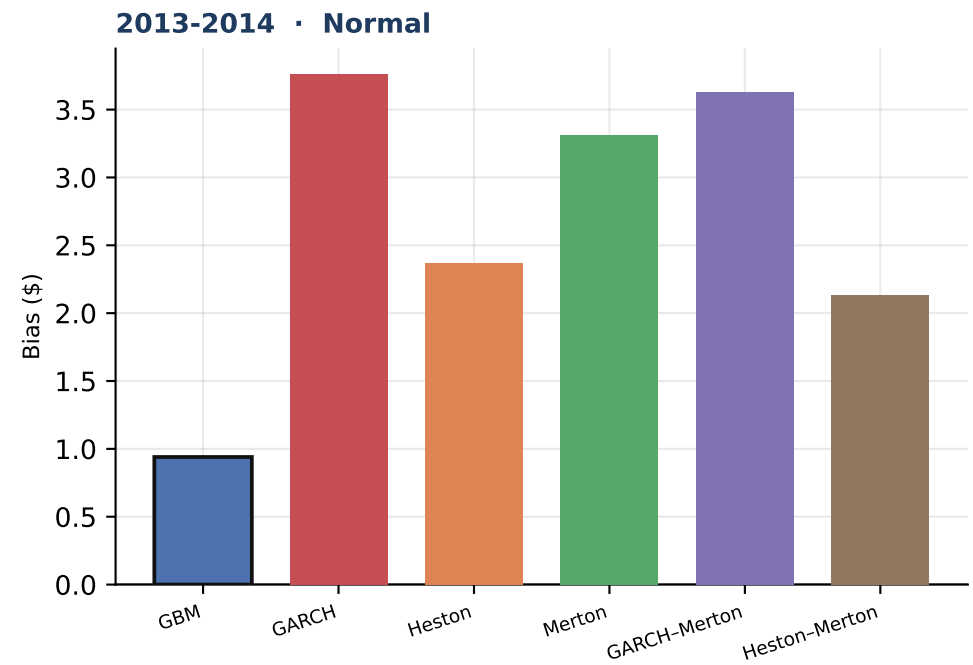
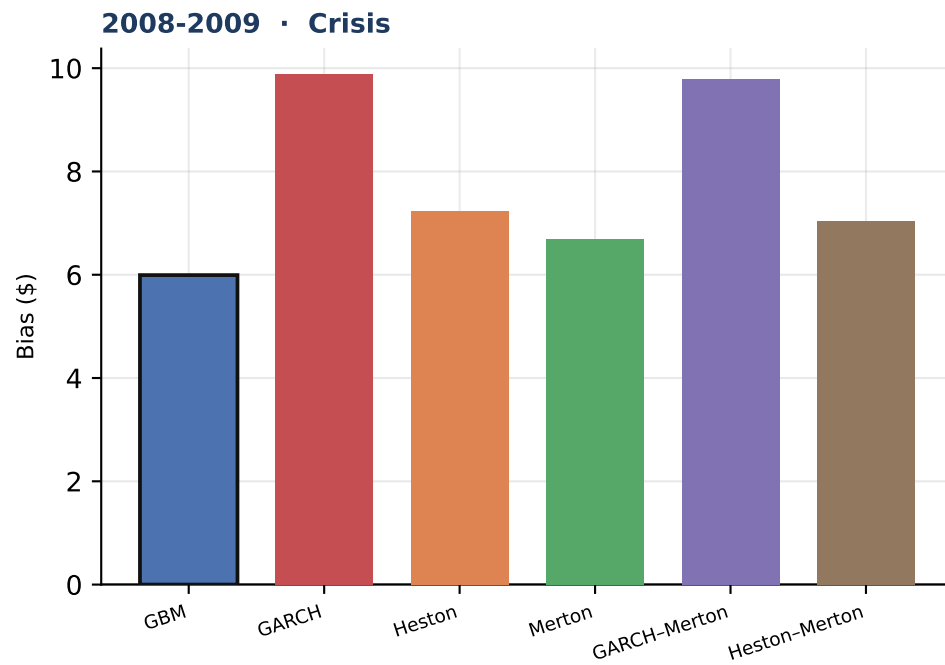
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the three companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Final analysis

Which models track realized S_t most closely?

- On mean RMSE% of the p50 path versus realized S_t across 12 company×regime cells, GBM is best (mean RMSE% = 18.35; 5 of 12 cells).
- The ranking is not stable. Best-in-cell models: GARCH-Merton, GBM, Heston, Merton.
- These conclusions are conditional on 1.5-year monthly P-measure calibration, 10000 paths, daily steps, and the p50 scoring convention.